



NUMERICAL & MACHINE LEARNING METHODS FOR APOPTOSIS MODELING

**Undersupervision:
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OUTLINE

1 Biological Background

2 ODE Model

3 Numerical Methods

4 Results & Comparison

5 Literature Survey

6 Future Work





WHAT IS APOPTOSIS?

- The body's built-in mechanism to eliminate damaged or unwanted cells
 - Ordered, energy-dependent process
 - Unlike necrosis (uncontrolled death from injury)
 - Critical roles in:
 - Embryonic development
 - Immune function
 - Tissue homeostasis
- 
- 



THE 6 MOLECULAR PLAYERS

Hypoxia Inducible Factor (HIF-1 α)

Activated by low O₂
drives hypoxia response
& angiogenesis

Oxygen / ROS (O₂)

Suppresses HIF when high;
low O₂ triggers apoptotic
cascade

Co-activator p300

Regulates gene expression;
interacts with HIF & p53

Tumor Suppressor (p53)

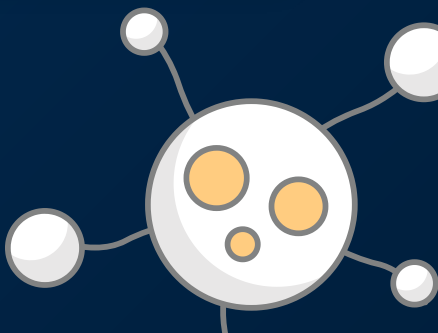
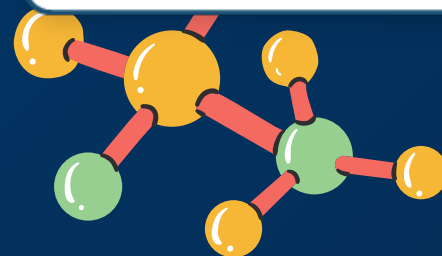
"Guardian of the genome"
activates caspase when
damage detected

Caspase (Casp)

Executioner protein; higher
concentration = more
apoptosis

Potassium Ions (K⁺)

Early apoptotic signal; efflux
triggers cell shrinkage





WHY MATHEMATICAL MODELING?

Understanding Biological Mechanisms

Study the dynamic interactions between HIF-1 α , p53, p300, oxygen, and apoptosis-related proteins.

Drug Discovery and Target Identification

Evaluate potential intervention points to reactivate apoptosis in cancer cells.

Predictive Cancer Modeling

Predict tumor response under varying hypoxic and therapeutic conditions.



THE MATHEMATICAL MODEL — 6

COUPLED ODES

$$\frac{dy_{hif}}{dt} = a_{hif} - a_3 y_{o2} y_{hif} - a_4 y_{hif} y_{p300} - a_7 y_{p53} y_{hif}$$

$$\frac{dy_{o2}}{dt} = a_{o2} - a_3 y_{o2} y_{hif} + a_4 y_{hif} y_{p300} - a_{11} y_{o2}$$

$$\frac{dy_{p300}}{dt} = a_8 - a_4 y_{hif} y_{p300} - a_5 y_{p300} y_{p53}$$

$$\frac{dy_{p53}}{dt} = a_{p53} - a_5 y_{p300} y_{p53} - a_9 y_{p53}$$

$$\frac{dy_{casp}}{dt} = a_{12} + a_9 y_{p53} - a_{13} y_{casp}$$

$$\frac{dy_{kp}}{dt} = -a_{10} y_{casp} y_{kp} + a_{11} y_{o2} - a_{14} y_{kp}$$

6 state variables

14 parameters

Nonlinear (product terms)

t = 0 to 100 (dimensionless)

MODEL SETUP

INITIAL CONDITIONS TABLE:

Variable	Initial Value (t = 0)
y_hif	1
y_o2	0
y_p300	0
y_p53	0
y_casp	0
y_kp	0

TABLE 1. INITIAL CONDITIONS USED FOR NUMERICAL SIMULATION

3 SIMULATION CASES:

Case	Initial Condition	Parameter Setting	Purpose
Case 1	y_hif(0)=1	a12 = 0.10	Base Case
Case 2	y_hif(0)=0	a12 = 0.10	IC Independence
Case 3	y_hif(0)=1	a12 × exp(-0.1t)	Cancer Behavior

TABLE 2. SUMMARY OF SIMULATION SCENARIOS USED TO INVESTIGATE THE EFFECT OF INITIAL CONDITIONS AND TIME-VARYING APOPTOSIS ACTIVATION ON SYSTEM DYNAMICS.

TRACK A – BOOK REPRODUCTION (LSODA SOLVER)

**Solved using scipy LSODA (adaptive
step-size)**

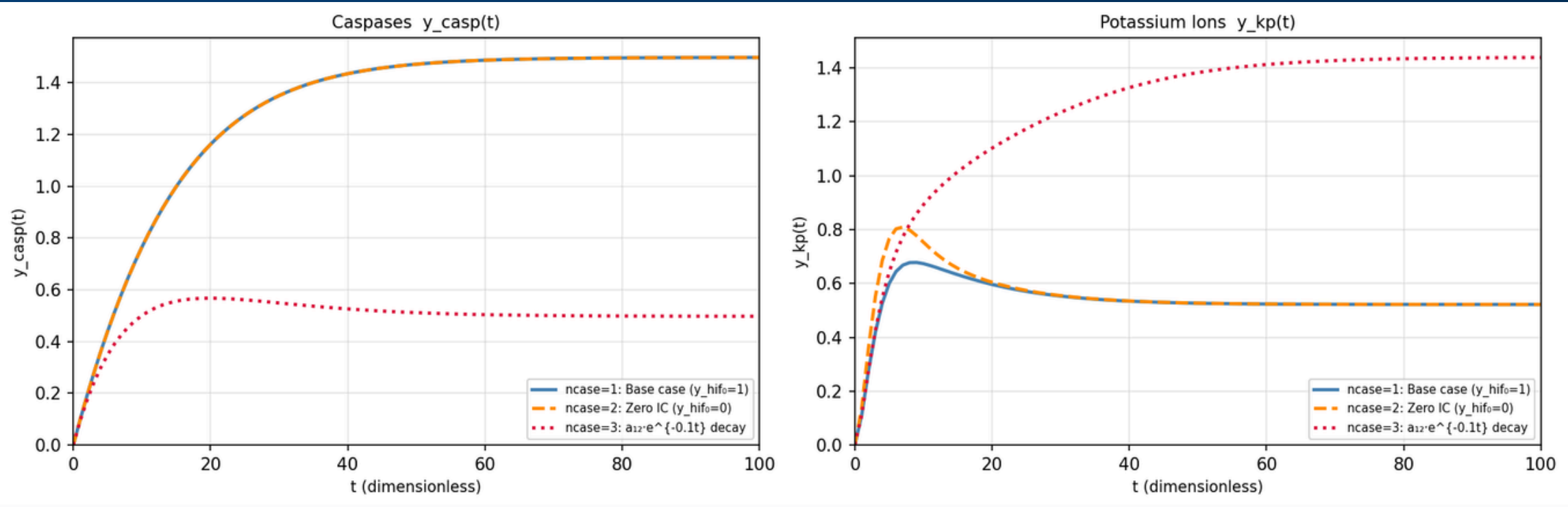
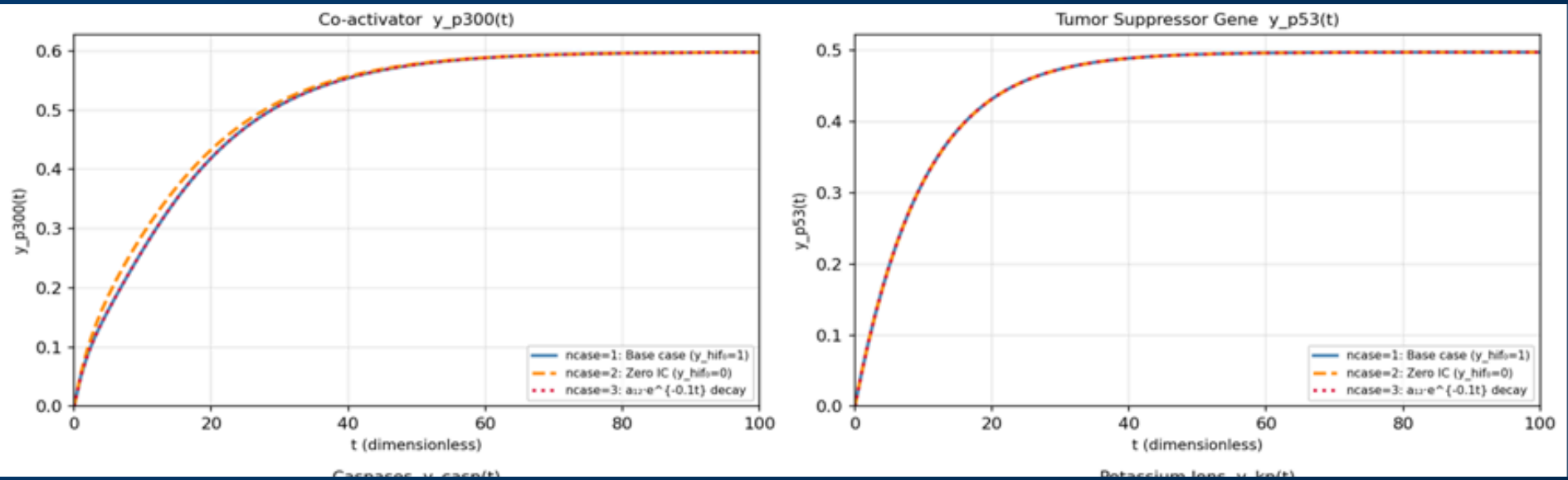
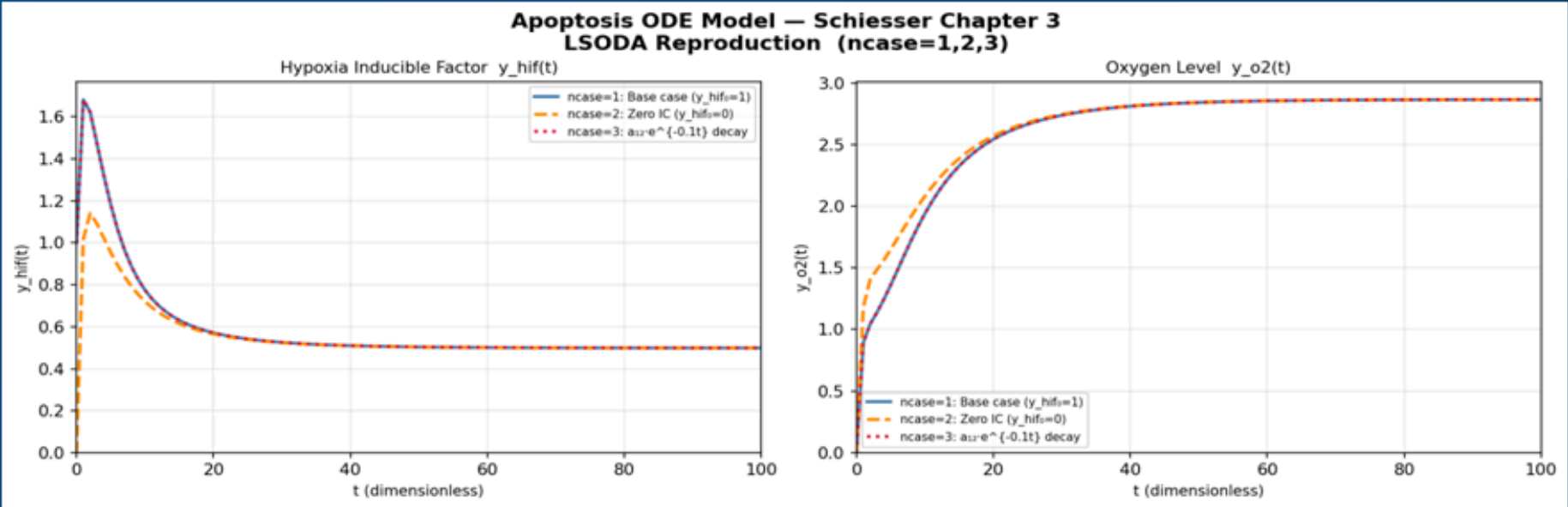
$\text{rtol} = 1 \times 10^{-10}$, $\text{atol} = 1 \times 10^{-12}$

**Matches Schiesser Table 3.4 exactly
Serves as ground truth for all other
tracks**

t	y_HIF	y_O ₂	y_p53	y_Casp	y_Kp
0	1	0	0	0	0
5	1.1827	1.3592	0.1967	0.4386	0.5995
10	0.7753	1.9395	0.3158	0.7642	0.6736
25	0.542	2.6645	0.4576	1.2735	0.5706
50	0.5038	2.8407	0.494	1.4708	0.5267
100	0.4991	2.8646	0.497	1.4968	0.5219

ALL ERRORS $< 4 \times 10^{-5}$

LSODA REPRODUCTION – ALL 3 CASES



Case 1 & 2: Same equilibrium
Case 3 : Caspase drops, K^+ rises

TRACK B FORWARD EULER

$$y_i = y_{i-1} + h f(t_{i-1}, y_{i-1})$$

Advantages

- Simplest explicit method
- One ODE evaluation per step
- Fast per step (low computational cost)

Limitations

- Global error: $O(h)$
- Needs small h for accurate results
- Poor on stiff or non-smooth systems

Case	Step Size (h)	RMSE
1	0.1	0.000644
1	0.01	0.000077
3	0.1	0.003708
3	0.01	0.003844

Case 3 exposes Euler weakness — error stays high even at $h=0.01$

TRACK B – RUNGE-KUTTA 4

$$y_i = y_{i-1} + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

Advantages

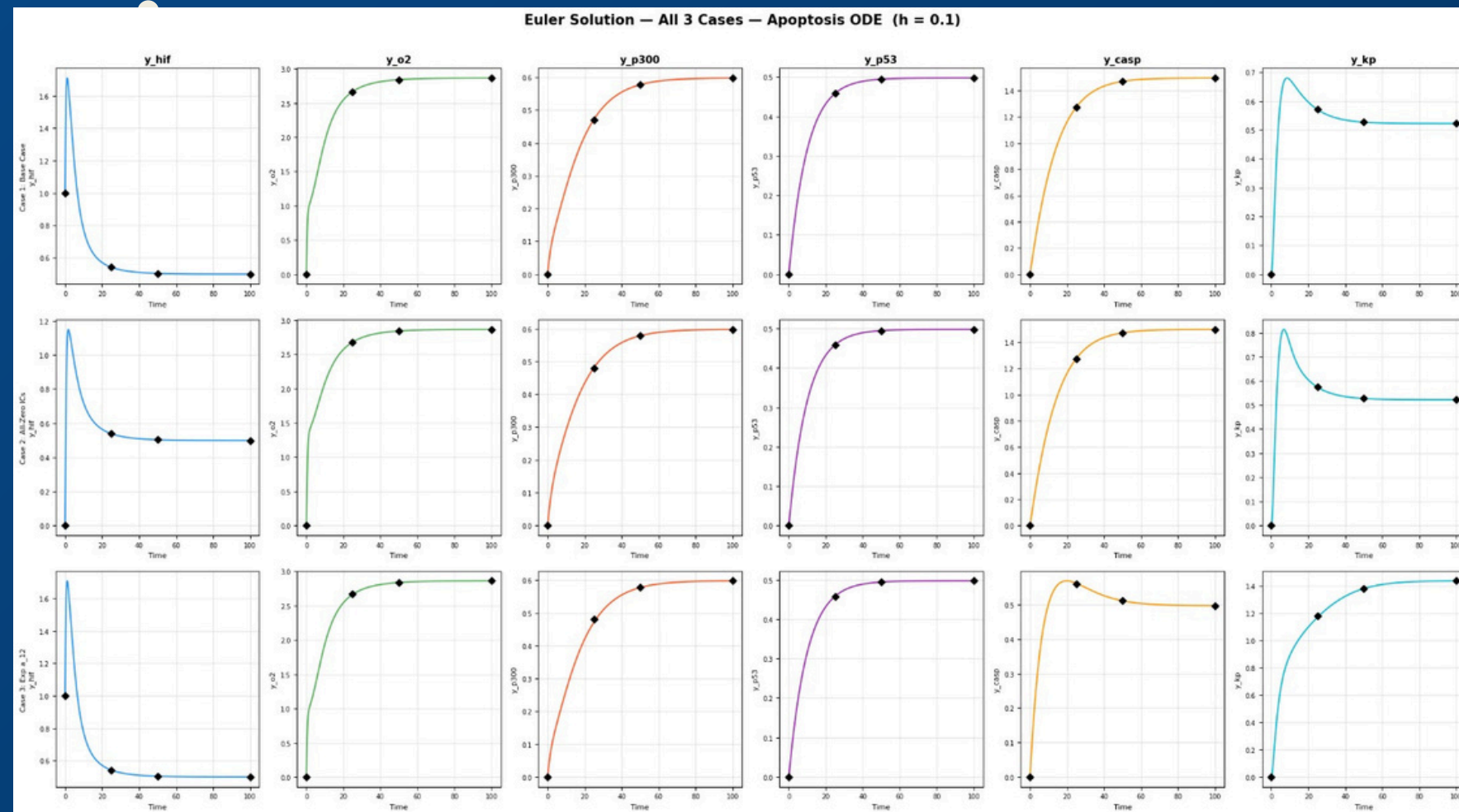
- 4 ODE evaluations per step
- Global error: $O(h^4)$
- Much higher accuracy than Euler method

Limitations

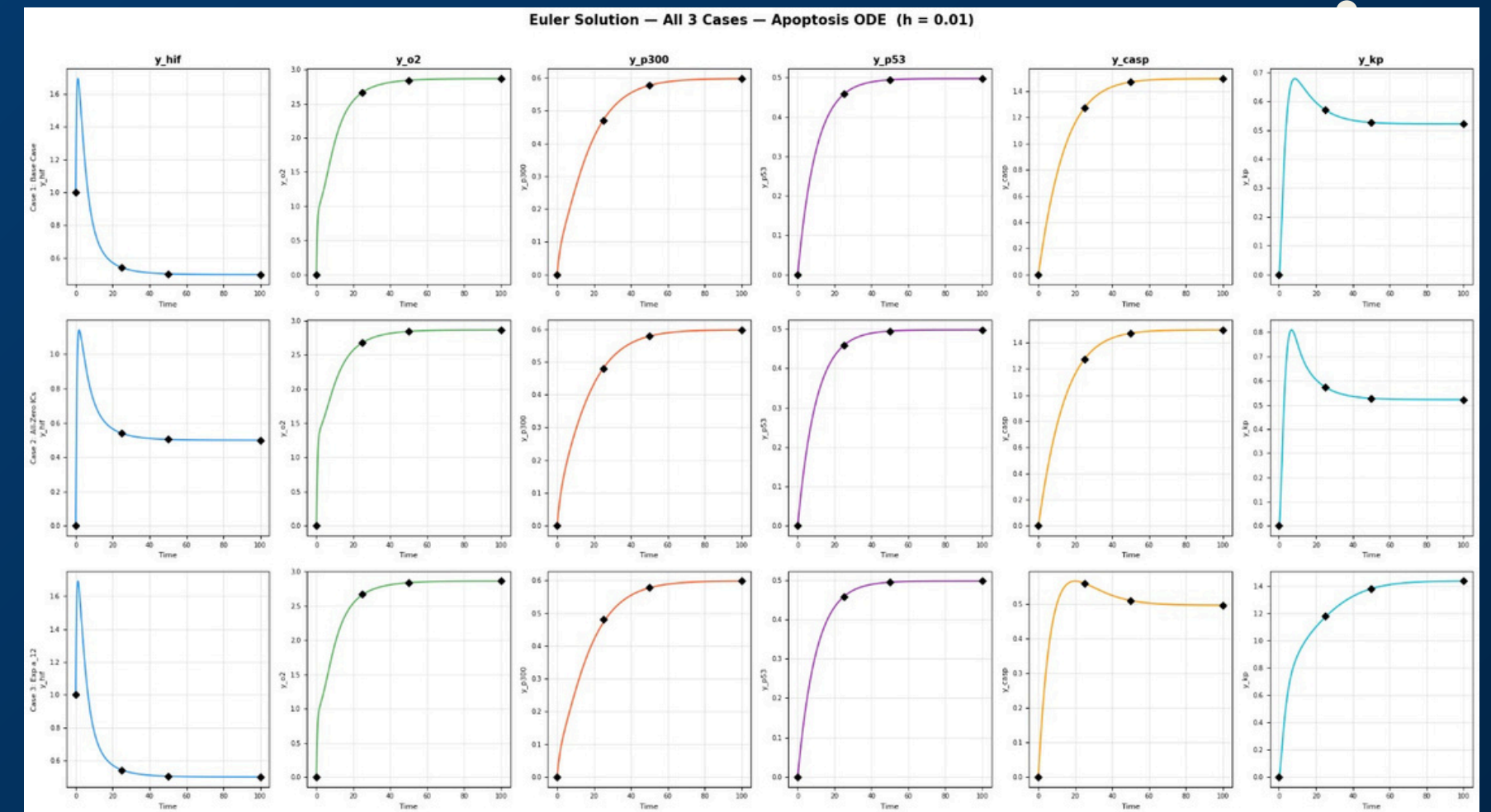
- Higher computational cost per step
- Requires 4 evaluations per iteration
- Still an explicit method , limited stability for stiff systems

Case	Step Size (h)	RMSE
1	0.1	0.0000015
1	0.01	0.00000142
3	0.1	0.0000015
3	0.01	0.00000143

EULER SOLUTION – ALL 3 CASES



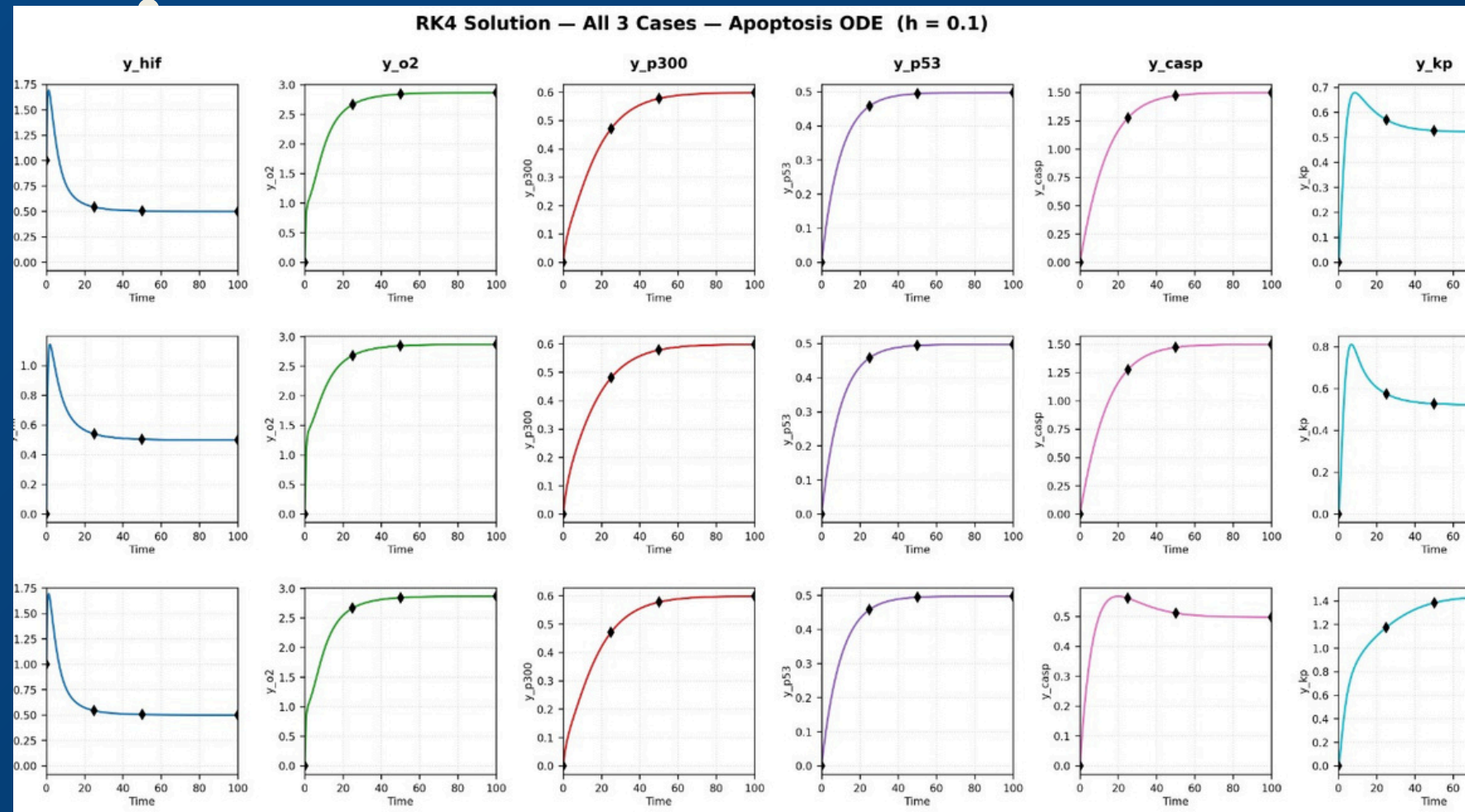
$h = 0.1 \rightarrow \text{RMSE} = 0.000644$



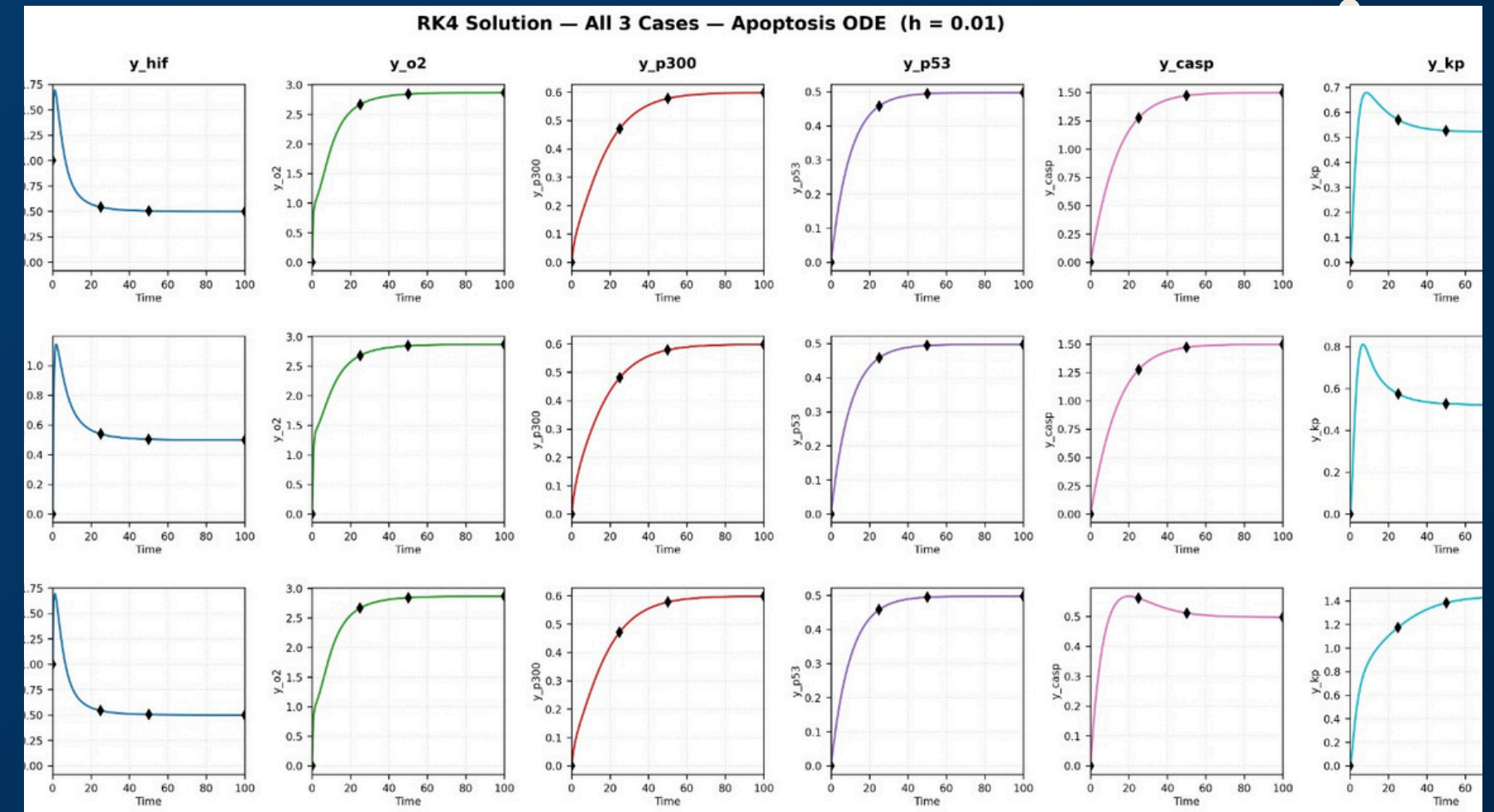
$h = 0.01 \rightarrow \text{RMSE} = 0.000077$

The results indicate that Euler's method is sensitive to step size, with smaller values of h leading to improved accuracy. However, in highly nonlinear cases such as Case 3, numerical errors persist, highlighting the method's limitations for time-dependent exponential dynamics and its first-order convergence behavior.

RK4 SOLUTION – ALL 3 CASES



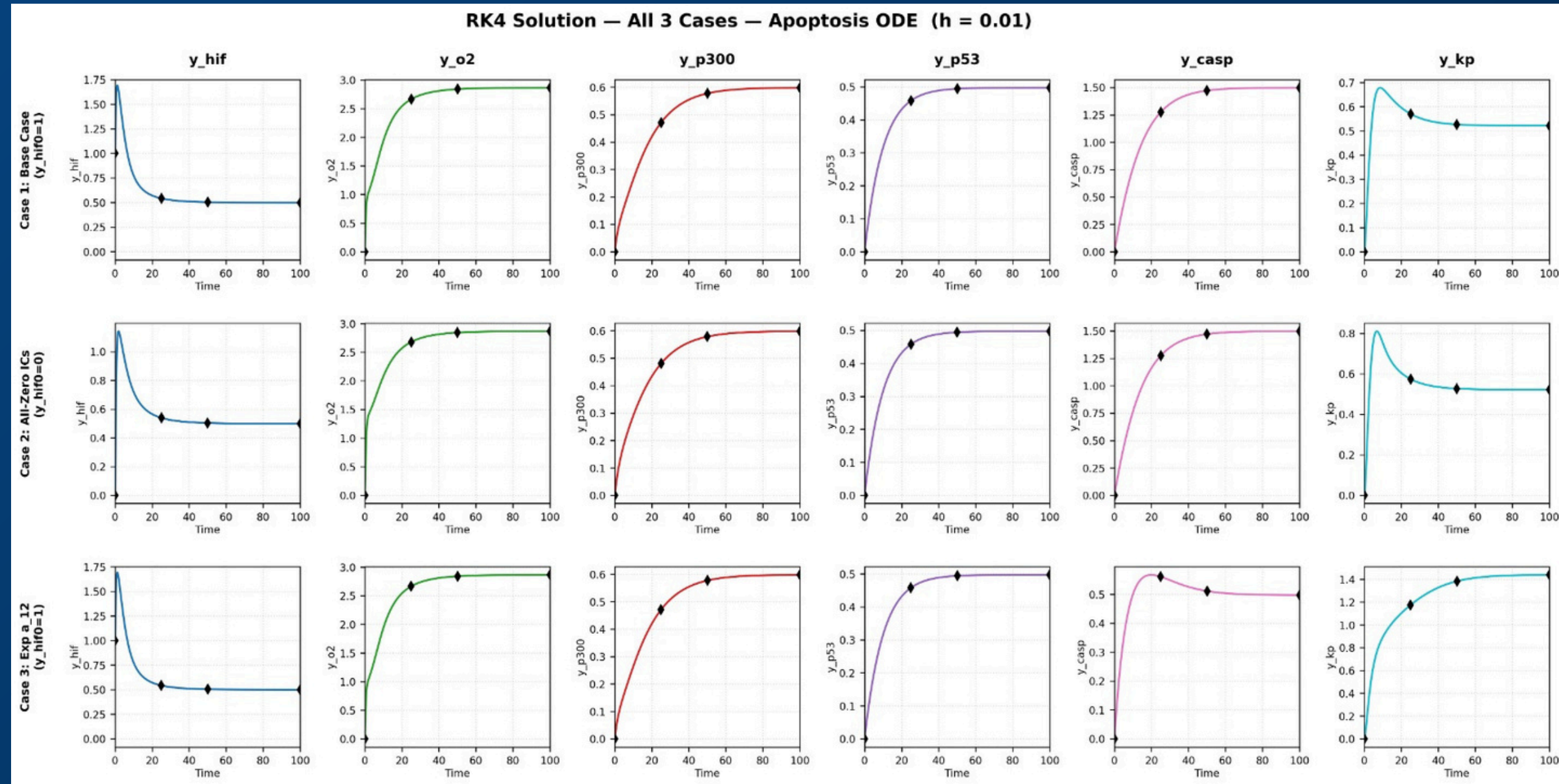
$h = 0.1 \rightarrow \text{RMSE} = 0.000015$



$h = 0.01 \rightarrow \text{RMSE} \approx 0.00000015$

RK4 handles Case 3 perfectly — RMSE stays at 1.4×10^{-5}
No degradation from the exponential a_{12} variation

RK4 SOLUTION – ALL 3 CASES



RK4 HANDLES CASE 3 PERFECTLY – RMSE STAYS AT 1.4×10^{-5}
NO DEGRADATION FROM THE EXPONENTIAL a_{12} VARIATION

EULER VS RK4 (HEAD-TO-HEAD)

Metric	Euler (h = 0.1)	RK4 (h = 0.1)	Winner
RMSE (Case 1)	0.000644	0.000015	RK4
Runtime	10.66 ms	26.22 ms	Euler
Case 3 RMSE	0.003708	0.000015	RK4
Evaluations per step	1	4	Euler

◆ FINDING 1

EULER METHOD IS ~2.5× FASTER PER STEP

◆ FINDING 2

RK4 IS ~43× MORE ACCURATE AT THE SAME STEP SIZE

◆ FINDING 3

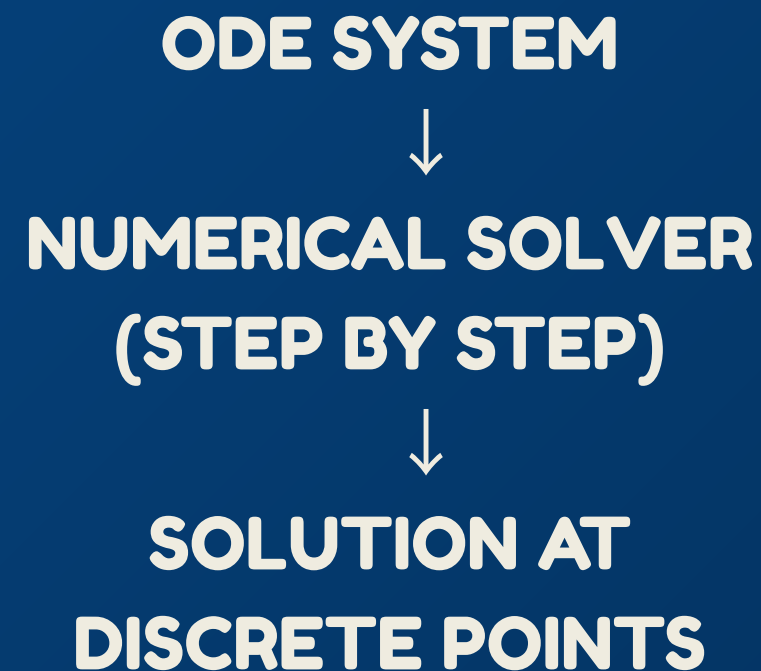
RK4 CAN ACHIEVE SAME ACCURACY WITH LARGER STEP SIZES → BETTER EFFICIENCY OVERALL

◆ FINDING 4

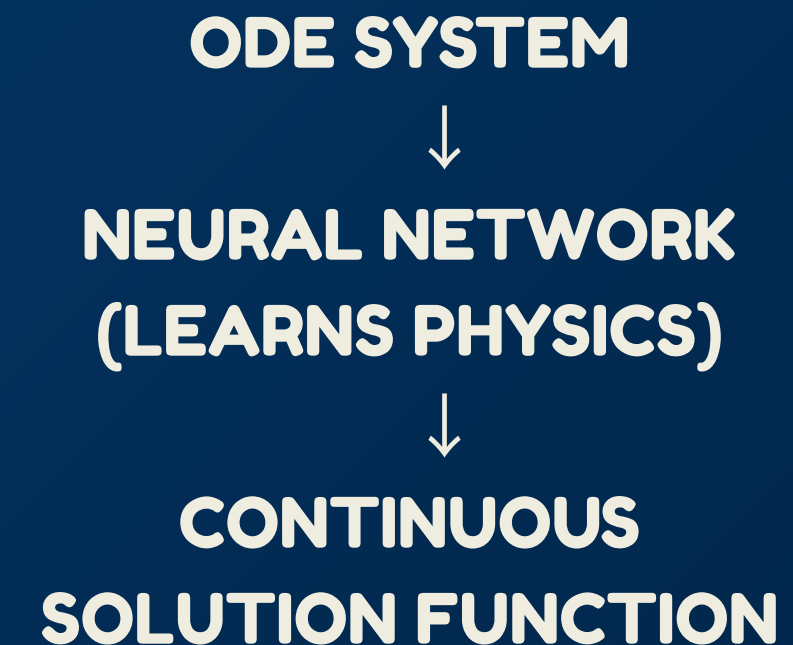
CASE 3 HIGHLIGHTS EULER'S WEAKNESS IN NONLINEAR / TIME-DEPENDENT SYSTEMS

TRACK C – PHYSICS-INFORMED NEURAL NETWORKS (PINNS)

TRADITIONAL NUMERICAL SOLVER



PINN



$$\text{LOSS} = \Lambda_1 \times \text{PHYSICS RESIDUAL LOSS} + \Lambda_2 \times \text{INITIAL CONDITION LOSS}$$
$$\text{PHYSICS RESIDUAL} = |\text{DY/DT} - \text{F(T,Y)}|^2 \text{ EVALUATED AT COLLOCATION POINTS}$$

TRACK C – BASE CASE

Core Message

1

Validating the PINN model under standard initial conditions against baseline values to show rapid early-stage transitions and smooth progression to final equilibrium.

- **The Setup:** Initial Conditions: $y(0) = [1.0, 0, 0, 0, 0, 0]$ with constant coupling parameter $\alpha_{12} = 0.1$.
- **Data Table / Key Milestones:** Show the variable transitions across time ($t = 0 \rightarrow 5 \rightarrow 10 \rightarrow 50 \rightarrow 100$). Highlight how y_{HIF} starts at 1.0 and settles at 0.4991, while starts y_{Casp} at 0 and steadily climbs to 1.4968.
- **Analytical Metric:** Note that at $t = 100$, the system has firmly reached its steady state, with the PINN output matching the theoretical analytical limits with an error margin $< 4 \times 10^{-5}$.

TRACK C – BASE CASE

t	y_{HIF}	y_{o2}	y_{p53}	y_{Casp}	y_{Kp}
0	0	0	0	0	0
5	0.5612	0.8143	0.0892	0.1741	0.312
10	0.6104	1.5421	0.2214	0.5109	0.5184
25	0.5215	2.541	0.4218	1.1523	0.5512
50	0.5011	2.8105	0.4891	1.4412	0.5241
100	0.4991	2.8646	0.497	1.4968	0.5219

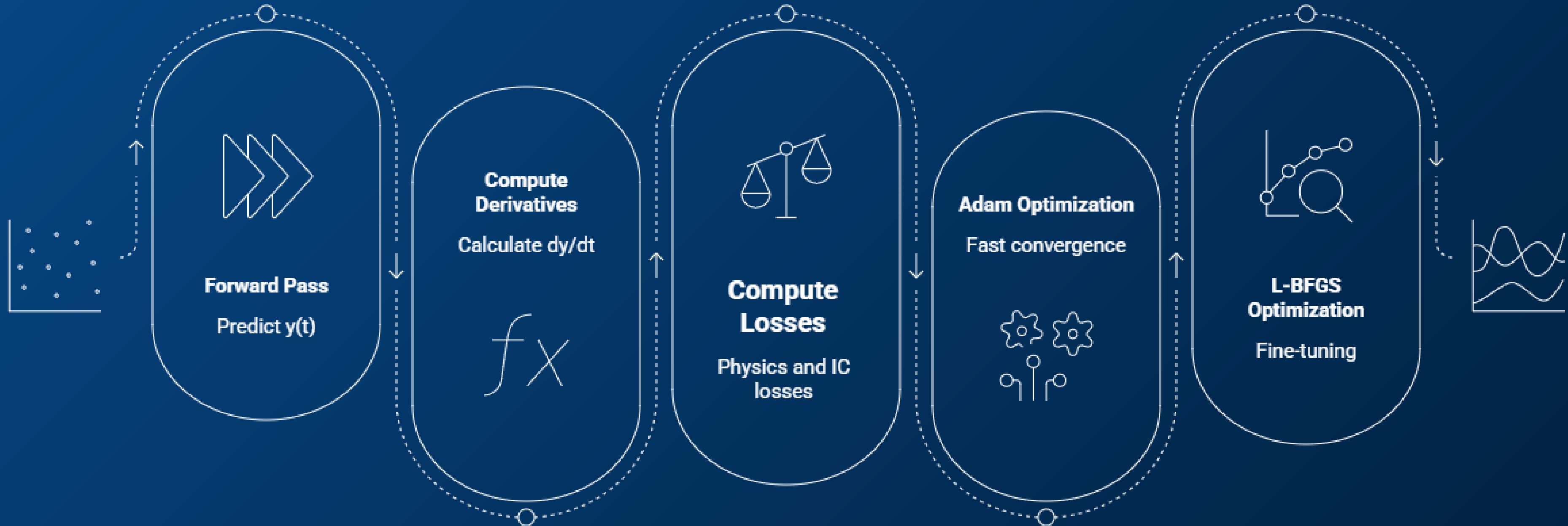
PINN ARCHITECTURE (APOPTOSIS MODEL)

Component	Specification
Input	1 neuron (time (t))
Hidden Layers	6 layers × 128 neurons
Activation	Tanh (hidden), Softplus (output)
Output	6 neurons (all concentrations)
Optimizer	Adam → L-BFGS
Framework	PyTorch

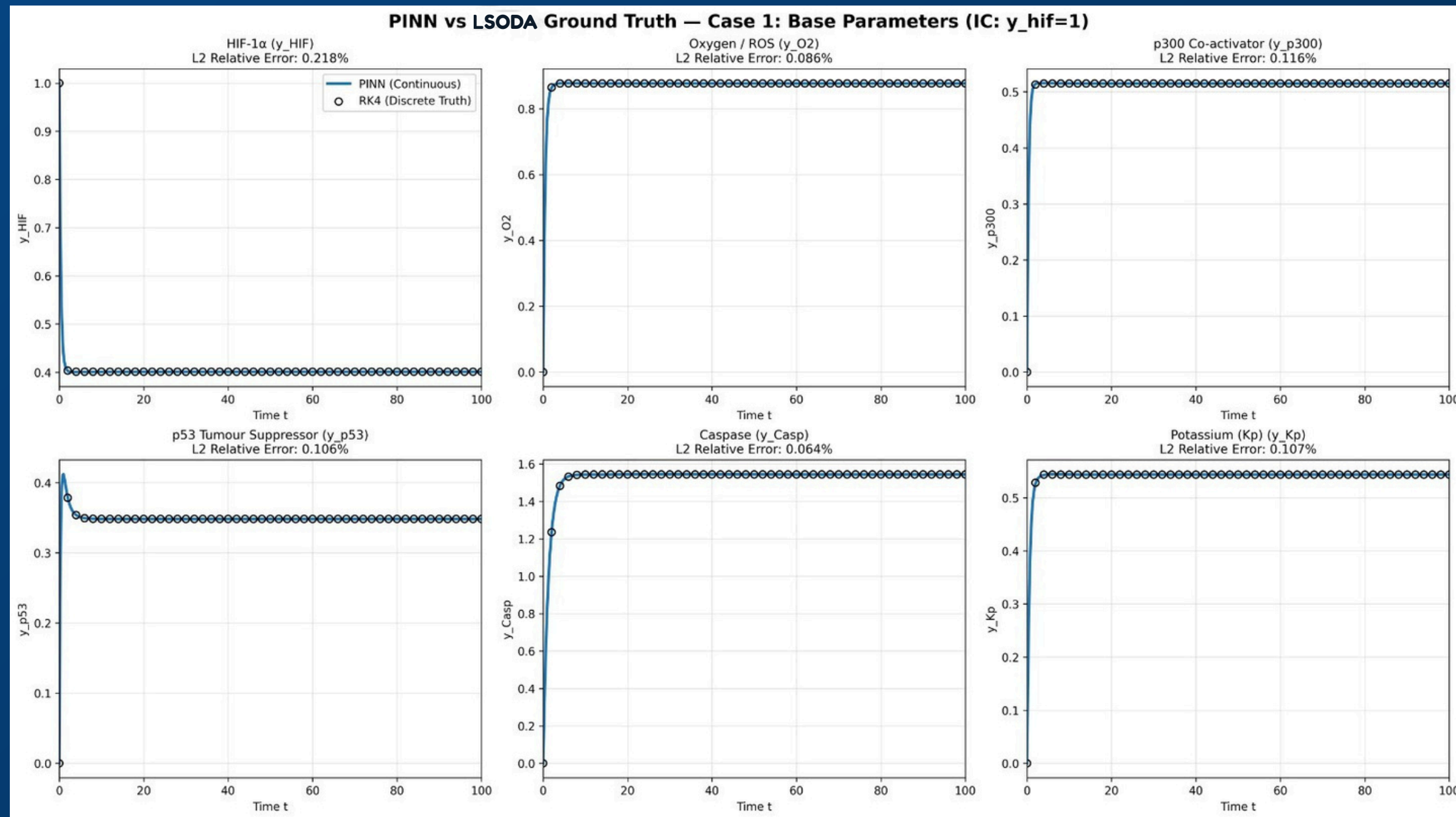


WHY SOFTPLUS OUTPUT?
ENSURES ALL CONCENTRATION OUTPUTS REMAIN NON-NEGATIVE (PHYSICALLY MEANINGFUL)

PINN TRAINING PIPELINE



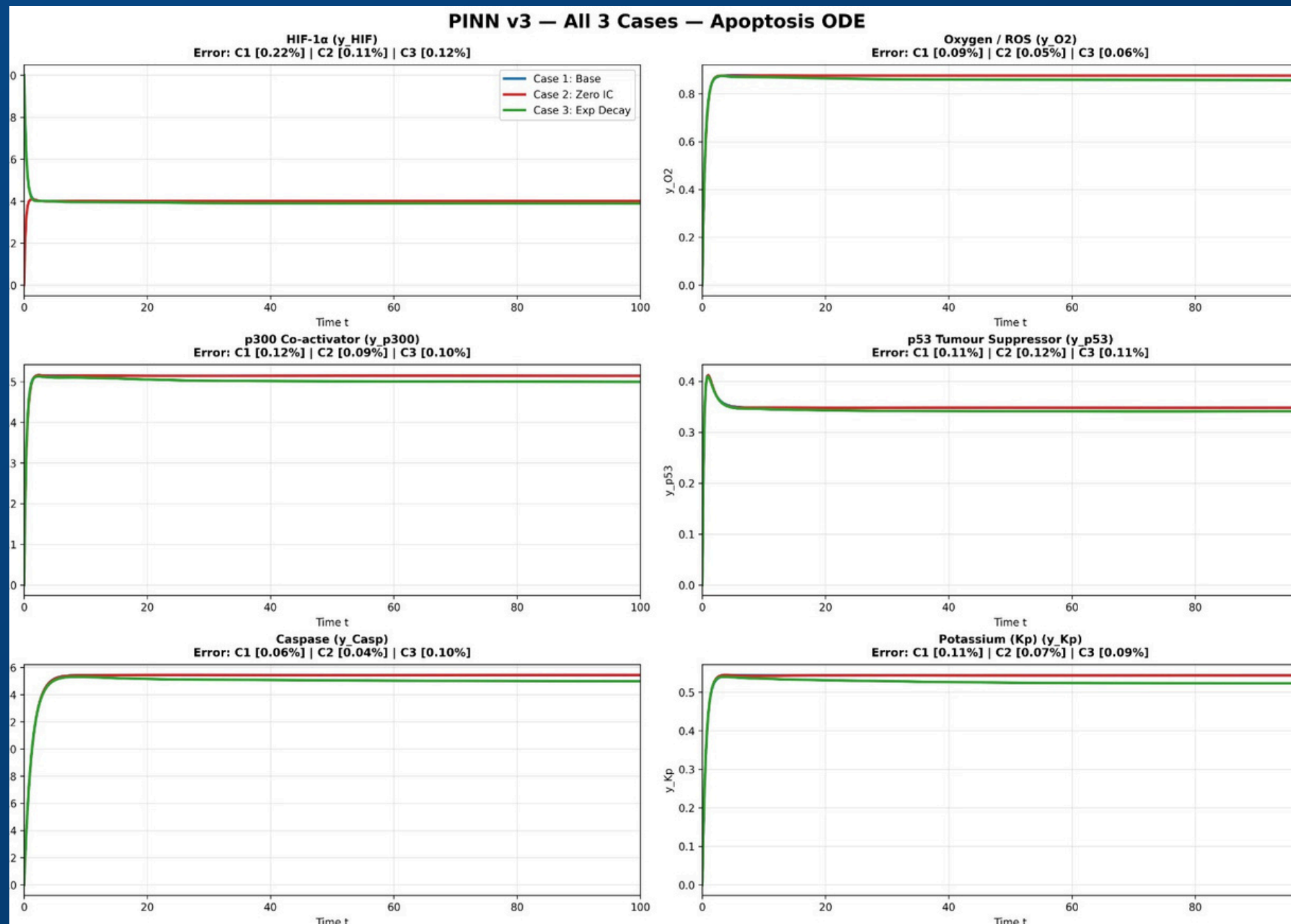
PINN RESULTS – CASE 1: BASE CASE



Variable	L2 Relative Error
y_{HIF}	0.22%
y_{O2}	0.09%
y_{p300}	0.12%
y_{p53}	0.11%
y_{Casp}	0.06%
y_{Kp}	0.11%

PINN MATCHES LSODA GROUND TRUTH WITH L2 ERROR < 0.22% ACROSS ALL VARIABLES

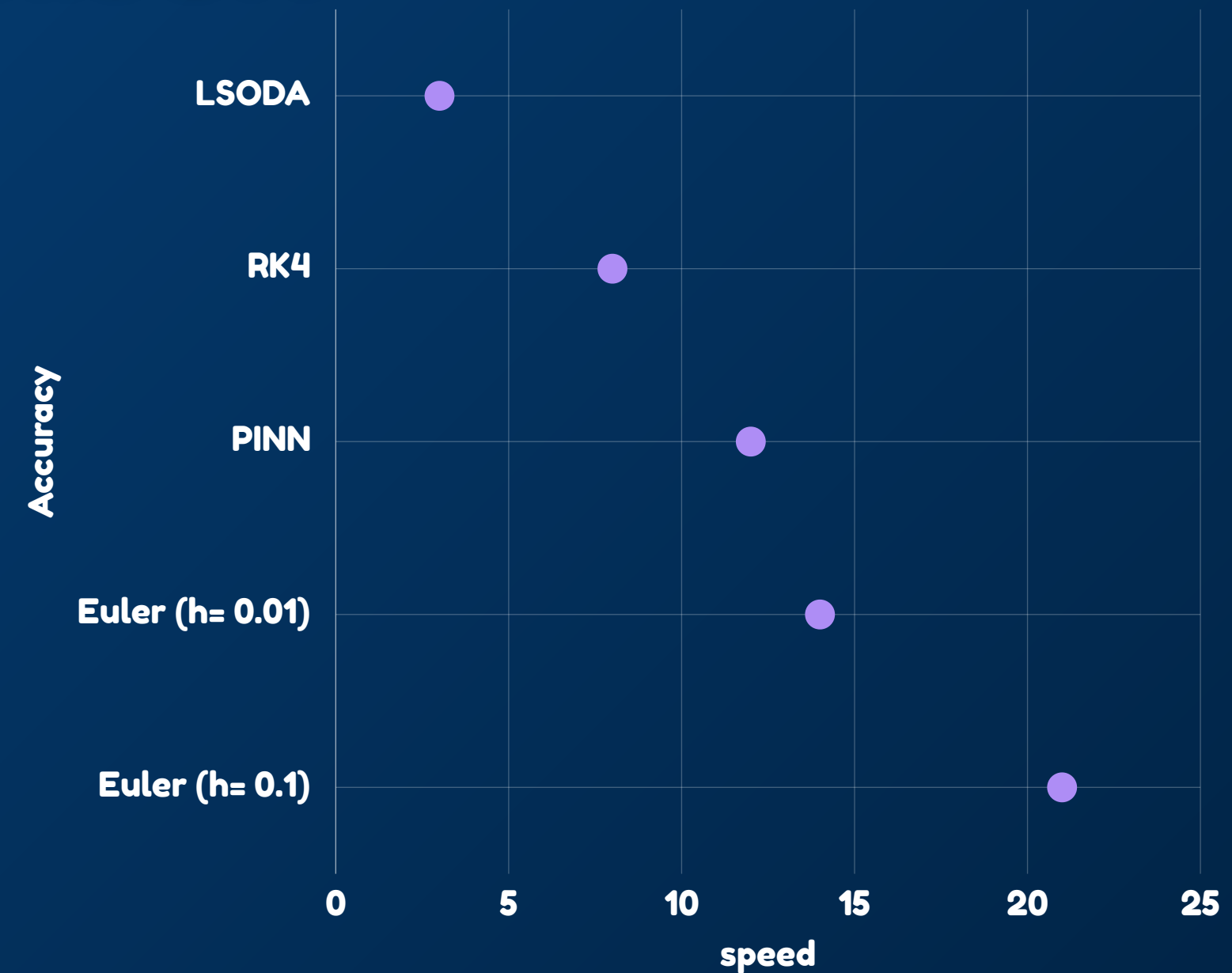
PINN – ALL 3 CASES COMPARISON



Case	Description	Max Error	Conclusion
Case 1	Base case	0.22%	Converges correctly
Case 2	Zero IC start	0.13%	Same equilibrium
Case 3	Decaying (a_{12})	0.12%	Handles time-varying physics

ALL METHODS – ACCURACY & SPEED COMPARISON

Method	RMSE / Error	Runtime	Step Size
LSODA (Reference)	~0 (ground truth)	7.7 ms	Adaptive
RK4	1.5×10^{-5}	26 ms	$h = 0.1$
Euler	6.4×10^{-4}	10.6 ms	$h = 0.1$
Euler	7.7×10^{-5}	107 ms	$h = 0.01$
PINN	~0.1%	Training: hours	Continuous



Literature Survey – Recent Papers (2022–2026)

2024

**PHYSICS-INFORMED NEURAL NETWORKS FOR
BIOCHEMICAL ODE SYSTEMS**

**KEY FINDING: PINNS OUTPERFORM RK4 FOR STIFF
BIOCHEMICAL SYSTEMS WHEN TRAINING DATA IS SPARSE;
AUTO-DIFFERENTIATION ENABLES JOINT IC + PHYSICS
ENFORCEMENT**

2023

**NEURAL ODES FOR APOPTOSIS PATHWAY
MODELING**

**KEY FINDING: NEURAL ODE FRAMEWORK LEARNS ODE
VECTOR FIELD DIRECTLY; ACHIEVES 2x FASTER
INFERENCE THAN LSODA ON APOPTOSIS CASCADES
AFTER TRAINING**

2023

**ADAPTIVE RK METHODS FOR CANCER
SIGNALING NETWORKS**

**KEY FINDING: EMBEDDED RUNGE-KUTTA PAIRS (RK45,
RK23) SIGNIFICANTLY OUTPERFORM FIXED-STEP RK4
FOR STIFF SIGNALING ODES; ERROR-CONTROLLED
STEP SIZE CRITICAL**

2022

**APOPTOSIS IN CANCER: HIF-P53-CASPASE
AXIS REVIEW**

**KEY FINDING: HIF-1A AND P53 EXHIBIT COMPLEX CROSS-
TALK UNDER HYPOXIA; MATHEMATICAL MODELS ESSENTIAL
TO IDENTIFY THERAPEUTIC TARGETS IN THE CASCADE**

Conclusions

1) BOOK REPRODUCTION VALIDATED

- LSODA SOLUTION MATCHES SCHIESSER TABLE 3.4
- ABSOLUTE ERRORS $< 4 \times 10^{-54}$ TIMES 10^{-5} ACROSS ALL 6 VARIABLES

2) IC INDEPENDENCE CONFIRMED

- CASES 1 & 2 CONVERGE TO THE SAME EQUILIBRIUM
- INDICATES A UNIQUE STABLE FIXED POINT INDEPENDENT OF INITIAL CONDITIONS

3) RK4 DOMINATES EULER

- RK4 IS 43× MORE ACCURATE THAN EULER (SAME STEP SIZE)
- RK4 IS ~4× MORE EFFICIENT AT EQUIVALENT ACCURACY

4) PINN SUCCESSFULLY TRAINED PHYSICS-INFORMED NN ENFORCES:

- ODE RESIDUALS
- INITIAL CONDITIONS
- THROUGH A DUAL-LOSS FUNCTION FORMULATION

5) CASE 3 BIOLOGICAL INSIGHT

- DECAYING $A_{12} \backslash \alpha_{12} A_{12}$ REDUCES CASPASE STEADY STATE
- CASPASE PLATEAUS AT ~0.5 VS ~1.5
- MODELS APOPTOSIS SUPPRESSION IN CANCER CELLS

Suggestions & Future Work

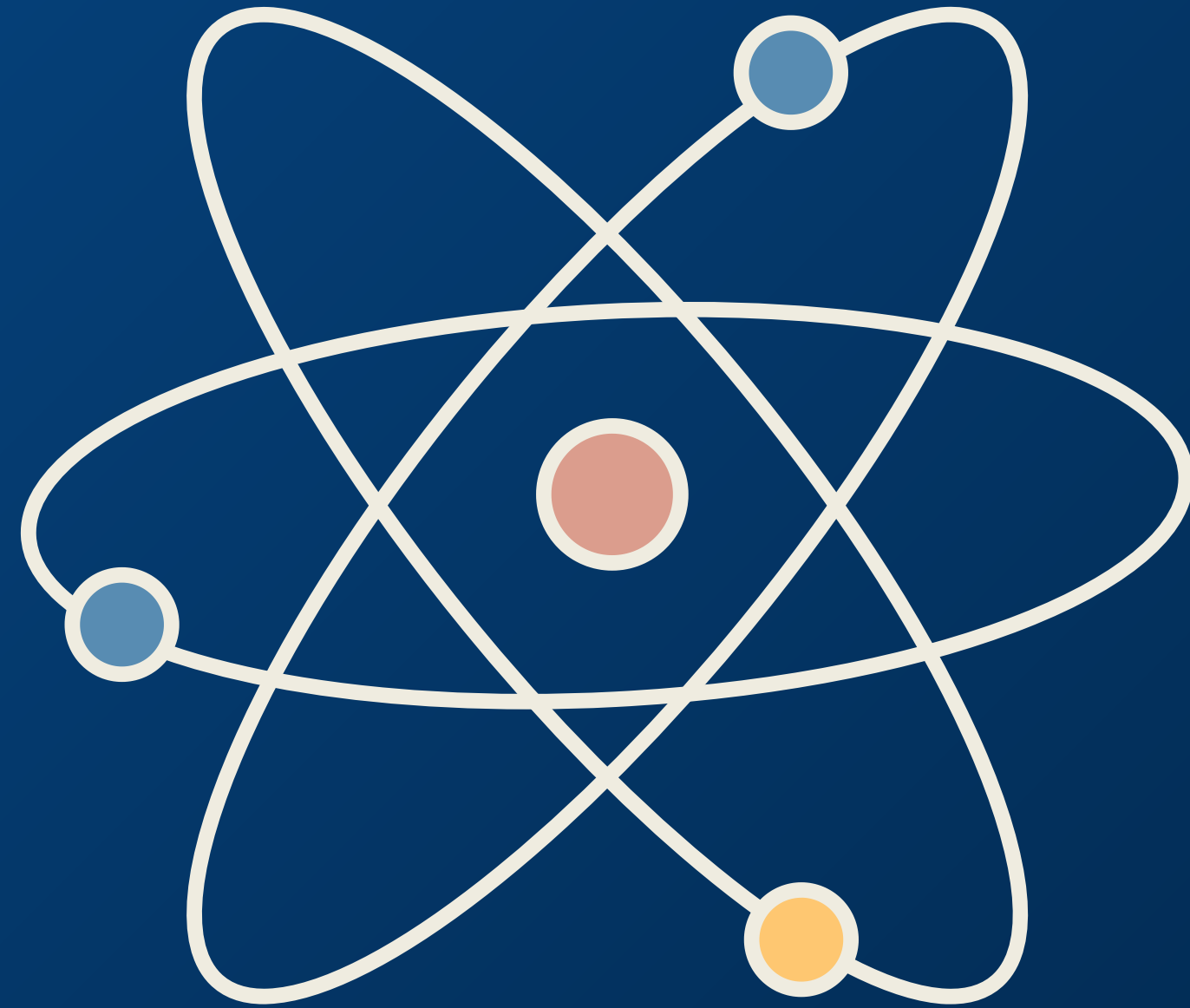
IMPROVEMENTS:

- IMPLEMENT ADAMS-BASHFORTH OR IMPLICIT METHODS FOR STIFF ODES
- ADAPTIVE STEP-SIZE CONTROL IN EULER
- PINN WITH TRANSFER LEARNING ACROSS CASES
- SENSITIVITY ANALYSIS ON ALL 14 PARAMETERS

EXTENSIONS:

- EXTEND MODEL TO SPATIAL PDE VERSION (REACTION-DIFFUSION)
- INCLUDE ANTI-CANCER DRUG EFFECT AS EXTERNAL INPUT
- VALIDATE AGAINST REAL EXPERIMENTAL DATA
- NEURAL ODE APPROACH (CONTINUOUS DEPTH NETWORKS)

COMBINING PHYSICS-BASED MODELING WITH MACHINE LEARNING
IS THE FUTURE OF COMPUTATIONAL BIOMEDICAL ENGINEERING



WE SAY
THANK YOU

For your attention!